

Design of Chebyshev I band-pass filter for 5G technology

The interface is opened and the design specifications are entered.

The 'Order' dialog box is shown with the following specifications:

- Ripple [dB]: 1
- Attenuation [dB]: 25
- ws/wp: 2.5

The 'Calculate filter order' button is visible. Below it, the filter type selection is shown:

- ☐ Butterworth (order 4)
- ☒ Chebyshev I (order 3)
- ☐ Chebyshev... (order 3)
- ☐ Elíptico (order 3)
- ☐ Pascal (order 3)

A 'Bessel Thomson' button is also present. At the bottom, the 'Transfer function' button is highlighted with a blue box, along with 'Bode of the approximations' and 'Finish' buttons.

Chebyshev I is selected and the transfer function is calculated.

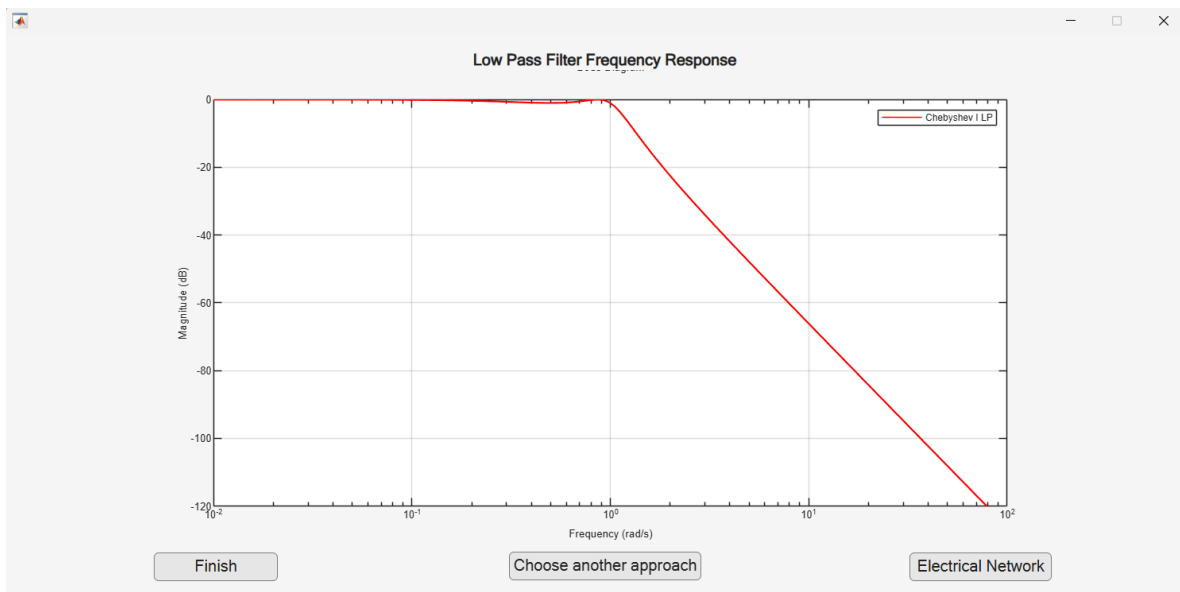
The 'Low Pass Filter Transfer Function' dialog box displays the calculated transfer function:

$$\text{sysch1_lp} = \frac{0.4913}{s^3 + 0.9883 s^2 + 1.238 s + 0.4913}$$

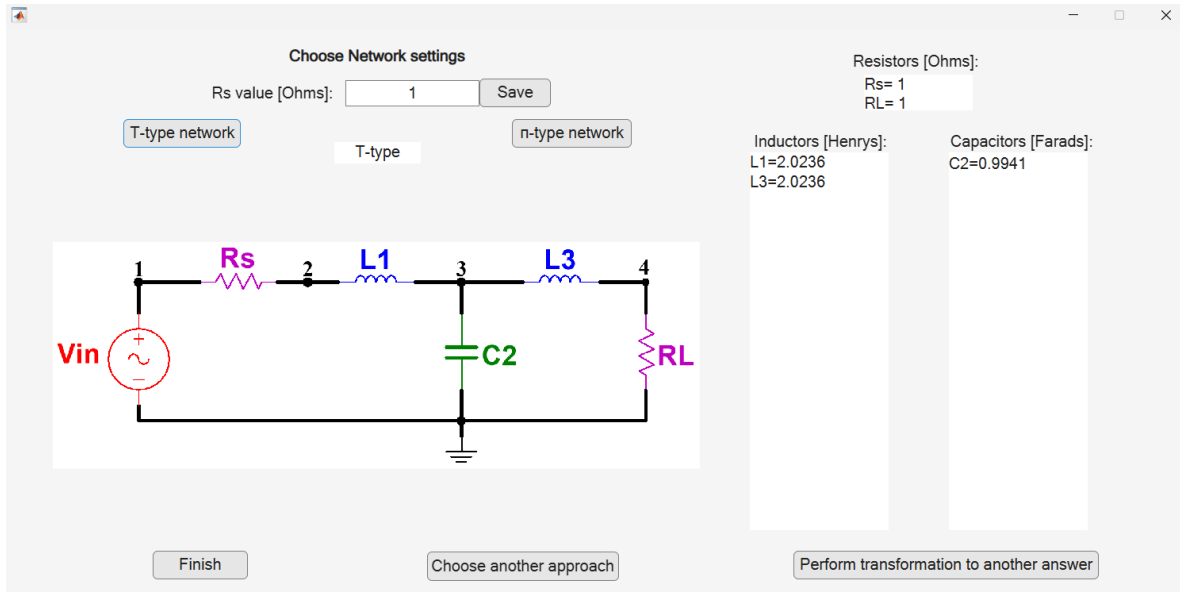
Below the equation, it states 'Continuous-time transfer function.' and shows a large block of MATLAB code for the transfer function, which is truncated in the image. At the bottom, there are 'Finish' and 'Frequency response' buttons.

The way of printing the transfer function in the most current versions of MATLAB numerically drags internal structures of the display.

Continuing, the magnitude frequency response is determined.



Then the normalized passive electrical network in T configuration.



The transformation from low-pass filter to band-pass filter is chosen.

Select transformation:

Lowpass filter impedance transformation

Lowpass filter frequency transformation

Lowpass to highpass filter transformation

Lowpass to bandpass filter transformation

Lowpass to bandreject filter transformation

Choose another approach method

Finish

The cutoff frequencies of 3.4 GHz and 3.79 GHz are entered to execute the transformation.

Enter Cutoff Frequency 1 [Hz]: 3.4E9

Enter Cutoff Frequency 2 [Hz]: 3.79E9

Transfer Function

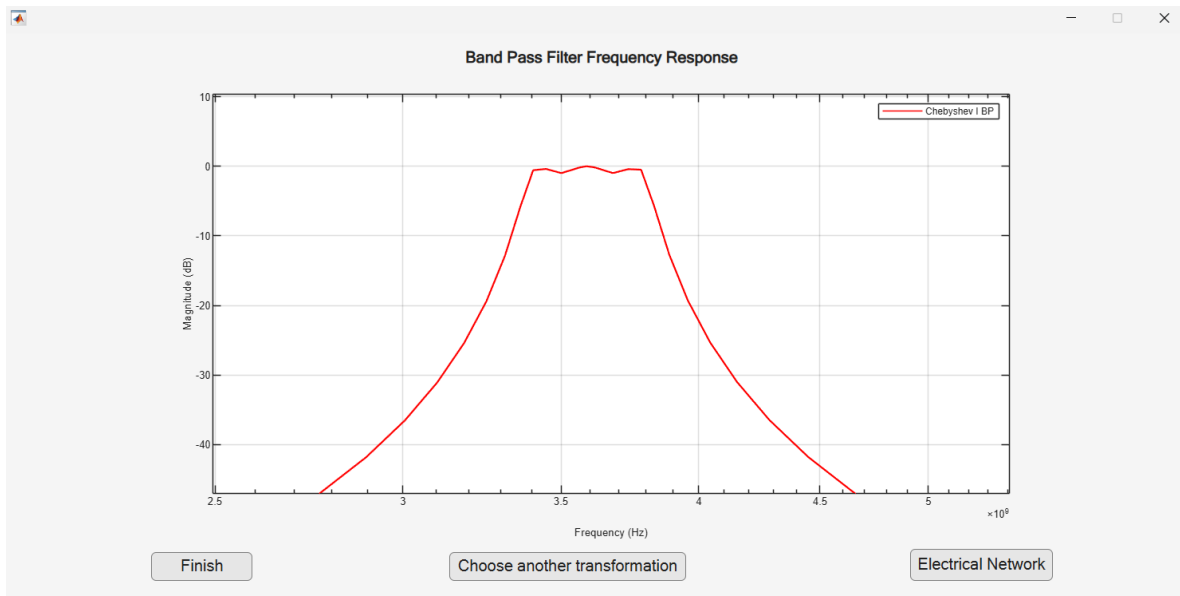
sysch1_bp_f =
$$\frac{7.229e27 s^3}{s^6 + 2.422e09 s^5 + 1.534e21 s^4 + 2.471e30 s^3 + 7.802e41 s^2 + 6.268e50 s + 1.317e62}$$

Continuous-time transfer function
matlab disp(char([10 32 32 32 32 32 78 117 109 101 114 97 116 111 114 58 32 123 91 48 32 48 32 55 46 50 50 57 49 101 43 50 55 32 48 32 48 32 48 93 125 10 32 32 32 32 32 68 101 110 111 109 105 110 97 116 111 114 58 32 123 91 49 32 50 46 52 50 49 57 101 43 48 57 32 49 46 53 51 51 54 101 43 50 49 32 50 46 52 55 49 51 101 43 51 48 32 55 46 56 48 49 55 101 43 52 49 32 54 46 50 54 55 55 101 43 53 48 32 49 46 51 49 54 53 101 43 54 50 93 125 10 32 32 32 32 32 32 32 86 97 114 105 97 98 108 101 58 32 39 115 39 10 32 32 32 32 32 32 73 79 68 101 108 97 121 58 32 48 10 32 32 32 32 32 73 110 112 117 116 68 101 108 97 121 58 32 48 10 32 32 32 32 32 79 117 116 112 117 116 68 101 108 97 121 58 32 48 10 32 32 32 32 32 32 73 110 112 117 116 78 97 109 101 58 32 123 39 39 125 10 32 32 32 32 32 73 110 112 117 116 85 110 105 116 58 32 123 39 39 125 10 32 32 32 32 32 73 110 112 117 116 71 114 111 117 112 58 32 91 49 215 48 32 115 116 114 117 99 116 93 10 32 32 32 32 32 79 117 116 112 117 116 78 97 108 101 58 32 123 39 39 125 10 32 32 32 32 32 79 117 116 112 117 116 85 110 105 116 58 32 123 39 39 125 10 32 32 32 32 32 79 117 116 112 117 116 71 114 111 117 112 58 32 91 49 215 48 32 115 116 114 117 99 116 93 10 32 32 32 32 32 32 85 115 101 114 68 97 116 97 58 32 91 93 10 32 32 32 32 32 32 32 32 32 32 32 32 78 97 32 78 111 116 101 115 58 32 91 48 215 49 32 115 116 114 105 110 103 93 10 32 32 32 32 32 32 84 115 58 32 48 10 32 32 32 32 32 32 84 105 109 101 85 110 105 116 58 32 39 115 101 99 111 110 100 115 39 10 32 32 109 101 58 32 39 10 32 32 32 32 32 32 32 32 32 32 32 32 84 115 58 32 48 10 32 32 32 32 32 32 84 105 109 101 85 110 105 116 58 32 39 115 101 99 111 110 100 115 39 10 32 32 32 32 83 97 109 112 108 105 110 103 71 114 105 100 58 32 91 49 215 48 32 115 116 114 117 99 116 93 10]))>Model Properties

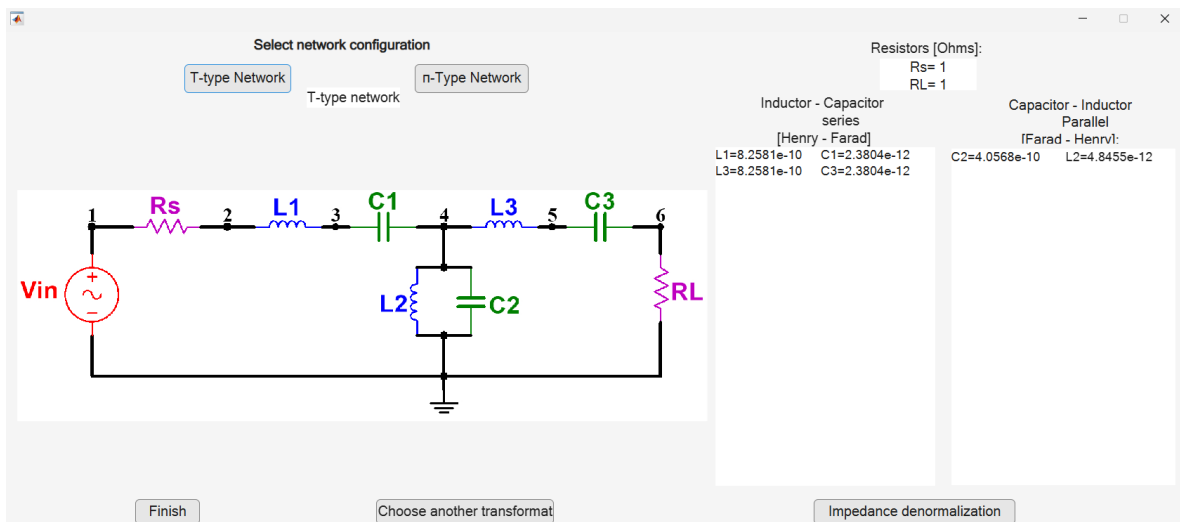
Finish

Frequency response

Subsequently the frequency response is generated.



The transformed electrical network is generated.



This result was recorded in Figure 9 of the article, as a normalized Chebyshev I band-pass transformed network.

Impedance denormalization is applied with a value of 50Ω.

Enter Denormalization Impedance [Ohms]:

50

Transfer Function

sysch1_bp_f =

$$\frac{7.229e27 s^3}{s^6 + 2.422e09 s^5 + 1.534e21 s^4 + 2.471e30 s^3 + 7.802e41 s^2 + 6.268e50 s + 1.317e62}$$

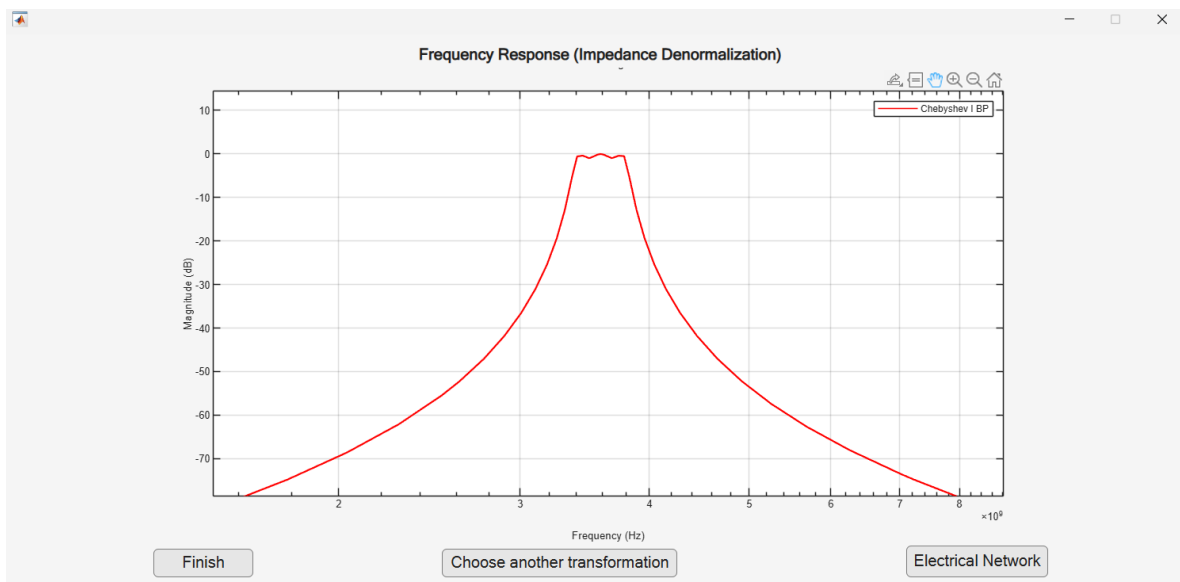
Continuous-time transfer function

Model Properties

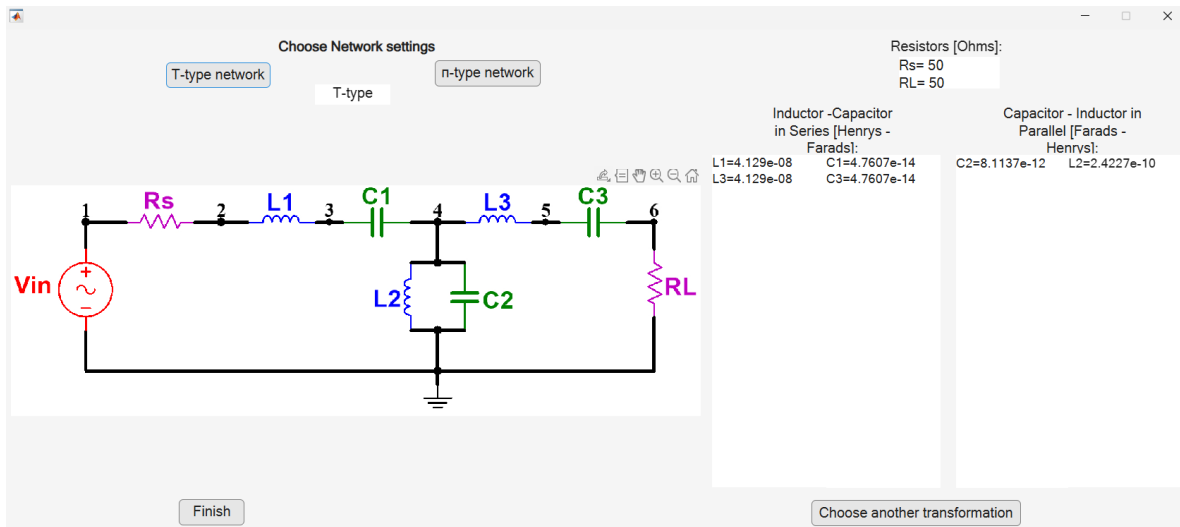
Finish

Frequency response

The transformed frequency response is visualized again.



Finally, the transformed and impedance denormalized electrical network is generated.



The externally obtained electrical network is subjected to the Richard transformation for implementation at high frequencies, giving rise to the Surface Integrated Waveguide shown in Figure 10 of the article.